Rigid Analytic Geometry Week 5 Solution

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Problem 1
If $x = 0$ or $y = 0$, then trivially the statement holds, thus we assume $x, y \in K^\times$. Then $xy^{-1} \in R$ or $yx^{-1} \in R$. This proves the statement.

Problem 2
$\mathfrak{m}_R = R \backslash R^\times$ is true by definition. Let $x \in R, y \in \mathfrak{m}_R$, if $xy \in R^\times$ then $y^{-1} = y^{-1}x^{-1}x \in R$ thus $xy \in \mathfrak{m}_R$. Note that $x^{-1}(x+y) \in \mathfrak{m}_R$, then by the previous argument $x + y \in \mathfrak{m}_R$. Thus without loss of generality, we assume $x|y$ in R. This is justified as R is a valuation ring. Then $y = rx$ for some $r \in R$. If $x + y$ is invertible, then

$$x(r+1)$$

is invertible. This shows both $x, r+1$ are invertible as $x, r+1 \in R$. We derived a contradiction and conclude $\mathfrak{m}_R$ is closed under addition.

Problem 3
The direction $\Leftarrow$ is clear as any proper ideal is contained in $R \backslash R^\times$. The other direction follows from that any non-unital element is contained in some maximal ideal thus $R \backslash R^\times \subseteq \mathfrak{m}$ where $\mathfrak{m}$ is the unique maximal ideal. As $\mathfrak{m}$ is proper, this is an equality.

Problem 4
Suppose for $x \in K$ be integral over R and $x \notin R$. Then $x^{-1} \in R$ and there are $a_1, \cdots, a_n \in R$ such that

$$x^n + a_1 x^{n-1} + \cdots + a_n = 0.$$

Then,

$$x = -(a_1 + a_2 x^{-1} + \cdots + a_n x^{n-1}).$$

All the elements in the right hand side are in R thus $x \in R$. Thus $x \in R$.

Problem 5
Let I, J be ideals of R. Suppose there are elements $a \in I, b \in J$ such that $a \notin J, b \notin I$. But either $a \in bR$ or $b \in aR$ thus this is not possible. I, J are linearly ordered so is the set of whole ideals.

Problem 6
Let $\{a_1, \cdots, a_n\} \subseteq R$ then this is linearly ordered by inclusion $(a) \subseteq (b)$ for some $a_i, a_j = a, b$. Let a be the maximal element, then,

$$(a_1, \cdots, a_n) = (a).$$

Problem 7

Prime ideals are equal to their own radicals and do not contain 1 as they are proper. If I is a proper ideal, then $1 \notin I$ thus $1 \notin \sqrt{I}$ otherwise it contains an unit. Let $ab \in \sqrt{I}$. Without loss of generality, we assume $b = ra$ for some $r \in R$. This is justified by the linearlity of ordering in ideals. Then,

$$b^2 = b \cdot ra \in \sqrt{I}.$$

Thus $b \in \sqrt{I}$ as $\sqrt{I}$ is invariant under taking radical.

Problem 8

As we have seen $\mathfrak{p}_{\mathfrak{U}}$ is a prime ideal, $R_{\mathfrak{U}}$ is a ring. Remains to show that this is a valuation ring of $\mathfrak{k}(\mathfrak{p})$. Let $z, n \in A \backslash \mathfrak{p}_{\mathfrak{U}}$ and $x = \frac{z}{n}$. Then we have,

$$V_A = \{(\mathfrak{p}, R) \mid v_{R,\mathfrak{p}}(x) \leq 1\} \cup \{(\mathfrak{p}, R) \mid v_{R,\mathfrak{p}}(x) \geq 1\}.$$

By the property of ultrafilters, one of the summands belongs to $\mathfrak{U}$. In particular, x or x^{-1} belong to $R_{\mathfrak{U}}$. Thus it is a valuation ring.

Finally, to show that it is a generic limit, enough to consider basis open elements,

$$\tilde{\mathcal{R}}(f_0|f_1),$$

we have,

$$(\mathfrak{p}_{\mathfrak{U}}, R_{\mathfrak{U}}) \in \tilde{\mathcal{R}}(f_0|f_1) \iff \tilde{\mathcal{R}}(f_0|f_1) \in \mathfrak{U}.$$

$(\mathfrak{p}_{\mathfrak{U}}, R_{\mathfrak{U}}) \in \tilde{\mathcal{R}}(f_0|f_1)$ if and only if

$$f_0 \notin \mathfrak{p}_{\mathfrak{U}}, f_1 \mod \mathfrak{p}_{\mathfrak{U}} \in f_0 \mod \mathfrak{p}_{\mathfrak{U}} R_{\mathfrak{U}}.$$

Thus $\tilde{\mathcal{R}}(f_0|f_1) \in \mathfrak{U}$ as for any $(\mathfrak{p}, R) \in \tilde{\mathcal{R}}(f_0|f_1)$,

$$v_{R,\mathfrak{p}}(f_1) \leq v_{R,\mathfrak{p}}(f_0).$$

Thus $(\mathfrak{p}_{\mathfrak{U}}, R_{\mathfrak{U}})$ is a generic limit. They are closed under glim as $\mathcal{R}(f_0|f_1, \cdots, f_n)$ are quasi-compact in Spv$A$.